Quadratics: The Discriminant

Questions 1–15

Question 1. Calculate the discriminant:

a) $x^2 - 2x - 6$
b) $x^2 - 9$
c) $-x^2 + x + 5$
d) $-0.5x^2 + 12$
e) $2x^2$
f) $2x^2 - 3x + 2$
g) $-3x^2 + 3x - 1$
h) $x^2 - kx + 2$

Question 2. State the number of roots for each of the following functions:

a) $f(x) = x^2 - 4x + 5$
b) $f(x) = x^2 + x - 6$
c) $f(x) = 2x^2 - 2x - 4$
d) $f(x) = x^2 - 3x + 8$
e) $f(x) = x^2 + 8x + 16$
f) $f(x) = -2x^2 + 5x + 3$
g) $f(x) = -4x^2 + 12x - 9$

Question 3. Show that $3x = 3x^2 + 1$ has no real solutions.

Question 4. Find the values of k for each of the following equations with equal roots:

a) $4x^2 + kx + 9 = 0$
b) $3x^2 + kx + 12 = 0$
c) $x^2 - 3x + 2 + \frac{1}{k} = 0$
d) $x^2 + (2k + 10)x + k^2 + 5 = 0$
e) $16x^2 + 4x + k^2 = 0$

Question 5. The equation $x^2 + 3px + p = 0$ where p is a non-zero, has equal roots. Find the value of p.

Question 6. Show that the discriminant of $(x - p)(x - q)$ is $(p - q)^2$.

Question 7. Show that the discriminant of $k(x-p)^2+q$ is $-4kq$.

Question 8. The curve $y=-\frac{1}{2}x^2+kx+k$ touches the line $2y-2x=5$ at one point. Find the possible value(s) of k.

Question 9. The line $5y + x = k$ touches the curve $2x^2 + 2y = 1$ once.

a) Find the value of k.

b) Hence find the values at which $5y + x = k$ intersects the x-axis and the y-axis.

Question 10. A function $f(x) = ax^2 + bx + c$ has constants a, b and c such that $a < 0$, $b > 0$ and $c < 0$. The discriminant of $f(x)$ is 0. Sketch the graph of $f(x)$. Label the intersections with the axes.

Question 11. Show that $y = x^2 + kx - x$ for $k > 0$, always has two real roots.

Question 12. For the equations $2x + y = 1$ and $x^2 - 4ky + 5k = 0$ where k is non-zero constant,

a) Show that $x^2 + 8kx + k = 0$.

b) Given that $x^2 + 8kx + k = 0$ has equal roots, find the value of k.

c) Hence for this value of k find the solution of the simultaneous equations.

Question 13. Find the relationship between non-zero constants p and q, such that the equation $y = px^2+(p+q)x+q$ has equal roots.

Question 14. The curves $f(x) = x^2 - kx - 4$ and $g(x) = -x^2 + x + 4k$ intersect once.

a) Find the possible value(s) of k.

b) Hence find the coordinates of the point(s) of intersection of the two curves.

Question 15. The equation $x^4 - 6x^2 + k = 0$ has two pairs of equal roots. Find the value of k.